EXTREMAL VECTORS AND NORM ATTAINMENT IN COMPLETE PICK SPACES

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Problem 1 (Bampouras–Brevig [1, p. 4, after Theorem 4])**.** What can be said about norm-attaining vectors and Hilbert points for the pair $(\mathcal{H}, \mathcal{H} \odot \mathcal{H})$ when $\mathcal{H}$ is a normalized complete Pick space?

Let $\mathcal{H}$ be a separable Hilbert function space on X with complete Pick kernel normalized at x_0, so $k_{x_0} = 1$ and $\|1\|_{\mathcal{H}} = 1$. Write $\mathcal{A} = \mathrm{Mult}(\mathcal{H})$. Inner products are linear in the first variable. The weak-product norm is the infimum of $\sum_j \|g_j\|_{\mathcal{H}} \|h_j\|_{\mathcal{H}}$ over representations $f = \sum_j g_j h_j$. A unit vector $f \in \mathcal{H}$ is called $\mathcal{H}$-extremal if

$$\langle qf, f \rangle_{\mathcal{H}} = q(x_0), \qquad q \in \mathcal{A}. \tag{1}$$

A nonzero $f \in \mathcal{H}$ is norm attaining if $\|f\|_{\mathcal{H}\odot\mathcal{H}} = \|f\|_{\mathcal{H}}$.

Theorem 1. *A nonzero vector $f \in \mathcal{H}$ is norm attaining if and only if $f/\|f\|_{\mathcal{H}}$ is $\mathcal{H}$-extremal.*

We use three complete Pick results. First, each $u \in \mathcal{H}$ has a representation

$$u = \frac{a}{1-b}, \qquad a, b \in \mathcal{A}, \quad \|M_b\| \le 1, \quad b(x_0) = 0. \tag{2}$$

This is the scalar case $k = s$ of [2, Theorem 1.1(i)], with scaling for arbitrary $\|u\|_{\mathcal{H}}$. Second, every $\mathcal{H}$-extremal vector is a contractive multiplier, by [2, Corollary 4.2(i)]. Third, every nonzero $F \in \mathcal{H} \odot \mathcal{H}$ admits exact factorization

$$F = gh, \qquad \|g\|_{\mathcal{H}} \|h\|_{\mathcal{H}} = \|F\|_{\mathcal{H}\odot\mathcal{H}}. \tag{3}$$

Indeed, in [3, Theorem 1.12] write $F = \varphi u^2$ and take $g = \varphi u$, $h = u$; the subinner/free outer pair has $\|\varphi u\|_{\mathcal{H}} = \|u\|_{\mathcal{H}}$.

Lemma 1 (Cancellation)**.** *Let $L \subseteq \mathcal{H}$ be a closed multiplier-invariant subspace. If $b \in \mathcal{A}$, $\|M_b\| \le 1$, $b(x_0) = 0$, and $(1-b)v \in L$ for $v \in \mathcal{H}$, then $v \in L$.*

Proof. For $0 < r < 1$, the Neumann series shows that $(1-rb)^{-1}$ is a multiplier, so

$$v_r = (1-rb)^{-1}(1-b)v \in L.$$

Putting $T = M_b$, the identity

$$(I - rT)^{-1}(I - T) = r^{-1}\big(I - (1-r)(I-rT)^{-1}\big)$$

and $\|(I-rT)^{-1}\| \le (1-r)^{-1}$ give $\|v_r\|_{\mathcal{H}} \le 4\|v\|_{\mathcal{H}}$ for $r \ge 1/2$. Positivity of the two-point multiplier kernel at x_0, x gives

$$\begin{pmatrix} 1 & 1 \\ 1 & (1-|b(x)|^2)k(x,x) \end{pmatrix} \ge 0, \qquad |b(x)|^2 \le 1 - k(x,x)^{-1} < 1.$$

Thus $v_r(x) \to v(x)$ at every $x \in X$. A weakly convergent subsequence as $r \uparrow 1$ has limit v, since point evaluations determine a vector in $\mathcal{H}$. The closed subspace L is weakly closed, so $v \in L$. □

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Set $\mathfrak{m} = \{q \in \mathcal{A} : q(x_0) = 0\}$ and, for $u \in \mathcal{H}$, set

$$M_u = \overline{\mathcal{A}u}, \qquad N_u = \overline{\mathfrak{m}u}.$$

Both are closed multiplier-invariant subspaces. Moreover, $N_u = \overline{\mathfrak{m}M_u}$, where the latter means the closed linear span of all such products. From Lemma 1, a representation Equation (2) gives

$$M_u = M_a, \qquad N_u = N_a. \tag{4}$$

Indeed, $a = (1-b)u$ gives one inclusion of the cyclic subspaces, and cancellation in M_a gives the reverse inclusion.

Lemma 2 (Products modulo the base-point ideal)**.** *Suppose $g, h \in \mathcal{H}$ and $f = gh \in \mathcal{H}$. If $u \in N_g$ and $uh \in \mathcal{H}$, then $uh \in N_f$.*

Proof. Choose representations

$$g = \frac{a}{1-b}, \qquad h = \frac{c}{1-d}, \qquad u = \frac{e}{1-t}$$

as in Equation (2). Since $(1-b)(1-d)f = ac$, two applications of Lemma 1 give $M_f = M_{ac}$ and hence $N_f = N_{ac}$. Also $N_g = N_a$ by Equation (4). Invariance gives $e = (1-t)u \in N_a$. Choose $q_j \in \mathfrak{m}$ with $q_j a \to e$ in $\mathcal{H}$. Multiplication by c is bounded, so $q_j ac \to ec$ and $ec \in N_{ac}$. Finally, $ec = (1-t)(1-d)uh$. Cancel $1-t$ and then $1-d$ in N_{ac}; the intermediate vector $(1-d)uh$ belongs to $\mathcal{H}$ because $uh \in \mathcal{H}$. This proves the claim. □

Proof of Theorem 1. Normalize $\|f\|_{\mathcal{H}} = 1$. First suppose f is $\mathcal{H}$-extremal, so $f \perp N_f$. Choose the exact factorization $f = gh$ from Equation (3). If $g \in N_g$, Lemma 2 would imply $f \in N_f$, a contradiction. The same argument interchanging g, h shows $h \notin N_h$.

Let

$$\gamma = \|P_{N_g^\perp} g\|_{\mathcal{H}} > 0, \qquad \varphi = \gamma^{-1} P_{N_g^\perp} g.$$

Then $\varphi \in M_g$, $\varphi \perp N_g$, and $\|\varphi\|_{\mathcal{H}} = 1$. For $q \in \mathfrak{m}$, we have $q\varphi \in N_g$, so $\langle q\varphi, \varphi\rangle = 0$. Subtracting $q(x_0)$ for general $q \in \mathcal{A}$ shows that φ is $\mathcal{H}$-extremal. In particular it is a contractive multiplier. Write

$$g = \gamma\varphi + u, \quad u \in N_g.$$

Since $uh = f - \gamma\varphi h \in \mathcal{H}$, Lemma 2 gives $uh \in N_f$. It follows that $\varphi h \in M_f$, and therefore

$$N_{\varphi h} \subseteq N_f.$$

Apply the same projection construction to h, obtaining

$$h = \delta\psi + v, \quad \delta = \|P_{N_h^\perp} h\|_{\mathcal{H}} > 0, \quad v \in N_h,$$

where ψ is a unit $\mathcal{H}$-extremal vector and a contractive multiplier. Apply Lemma 2 to the pair (h, φ) and to v. Both $h\varphi$ and $v\varphi$ belong to $\mathcal{H}$, because φ is a multiplier. Thus $\varphi v \in N_{\varphi h} \subseteq N_f$. We have proved

$$f - \gamma\delta\varphi\psi = uh + \gamma\varphi v \in N_f.$$

Pairing with $f \perp N_f$ and using $\|\varphi\psi\|_{\mathcal{H}} \le \|M_\varphi\|\|\psi\|_{\mathcal{H}} \le 1$ yields

$$1 = \gamma\delta|\langle\varphi\psi, f\rangle| \le \gamma\delta \le \|g\|_{\mathcal{H}}\|h\|_{\mathcal{H}} = \|f\|_{\mathcal{H}\odot\mathcal{H}}.$$

The factorization $f = f \cdot 1$ gives the reverse inequality.

Conversely, suppose f is norm attaining. By [1, Theorem 1], its coefficient functional $L_f(u) = \langle u, f\rangle_{\mathcal{H}}$ extends to $\mathcal{H} \odot \mathcal{H}$ with norm 1. Let A be the conjugate-linear Hankel operator specified by $\langle g, Ah\rangle = L_f(gh)$. Then $A1 = f$ and $\|A\| = 1$. For $q \in \mathcal{A}$ with $q(x_0) = 0$, one has $q \perp 1$. Expanding $\|A(1+zq)\|^2 \le \|1+zq\|^2$ for arbitrary complex z shows $\langle f, Aq\rangle = 0$. Hence $\langle fq, f\rangle = 0$. Subtracting the constant value $q(x_0)$ proves Equation (1) for every multiplier. □

Corollary 1. *Let f be a unit $\mathcal{H}$-extremal vector and $M = M_f$. For every $h \in \mathcal{A}$,*

$$\|P_{M^\perp} M_h^* f\|_{\mathcal{H}}^2 \le \|h\|_{\mathcal{H}}^2 - |h(x_0)|^2. \tag{5}$$

This applies to Drury–Arveson space H_d^2, $1 \le d \le \infty$.

Proof. By Theorem 1 and [1, Theorem 1], the Hankel form of f has norm 1. Therefore $\|M_h^* f\|_{\mathcal{H}} \leq \|h\|_{\mathcal{H}}$. Extremality gives, for every $q \in \mathcal{A}$,

$$\langle qf, M_h^* f\rangle = \langle hqf, f\rangle = h(x_0)q(x_0) = \langle qf, \overline{h(x_0)}f\rangle.$$

Taking the closure yields $P_M M_h^* f = \overline{h(x_0)}f$. Pythagoras now proves Equation (5). □

Remark 1. The theorem answers the norm-attaining part of the source question in the stated separable setting, including $f(x_0) = 0$. These vectors are Hilbert points by [1, Corollary 3]; classification of all Hilbert points is not asserted. Extremal and subinner remain distinct notions in [3, Definitions 1.1–1.2].

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References

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